

Project: Azure Capacity Reservation Management Engine (ACRME)

Classification: Principal Cloud Architect - Architecture Governance

Version: 2.3

Date: 7 September 2026

Status: Accepted - supersedes ADR-001 v2.2 region-selection content

Part of: ACRME Architecture Decision Records - aligned to Capacity & Quota Management Requirements Baseline v2.3.

About ADRs. An Architecture Decision Record captures a significant architectural decision, the context that forced it, the options considered, the choice made, and its consequences. This v2.3 ADR updates the accepted region-selection decision to match the consolidated requirements baseline — notably the five-geography region model with per-geography distribution models and mandatory two-region CVAL/DR co-location (PLC-010a). Evidence tags: [Documented] , [Decided] , [Derived] , [Assumed] .

ADR-001 - Region Selection and Customer Placement

Status: Accepted

Date: 27 August 2026

Deciders: Principal Cloud Architect, Platform Engineering, DR Owner, Product Platform

Related requirements: REG-001..REG-005, PLC-001..PLC-010, RDY-001..RDY-004, DR-014, DR-016..DR-019, DAT-002, DAT-005

Related constraints: ENV-003, CAP-011, CAP-012, QUA-012, NFR-002, NFR-010

Context

Requirements v2.1 changes the placement model from repeated geography-driven selection to a seeded, production-region-first model. Customers and product teams must know the exact Azure production region up front because a broad geography selection creates contract churn, data-residency ambiguity, and inconsistent product placement for the same customer. [Derived]

ACRME must still calculate CVAL and DR placement using current capacity, reservation, quota, zone, sharing, restriction, and DR-readiness state. The first valid decision becomes an authoritative seed that future products reuse. [Decided]

Key forces:

- Production protection is the primary objective. [Decided]
- Region, zone, quota, and capacity are hard isolation boundaries; capacity is never counted across regions or zones. [Documented]
- Region strategy is configuration-driven and spans **five supported geographies** (US, EU, Australia, Asia Pacific, Middle East), each catalogued with an explicit **distribution model** (baseline Section 6, REG-001). The US is the only three-region geography; EU, Australia, Asia Pacific and the Middle East are two-region geographies. New regions (e.g. Japan East, pending) are policy data added by config, not automatic rollout commitments. [Derived]

- **Middle East DR is currently `DR_NOT_OFFERED` (DR-014, DEC-001 — under legal review).** As it stands, DR is **not offered** in the Middle East: Legal owns the programme and data-sovereignty/residency laws (a largely government/medical customer base) mean cross-border DR cannot meet residency requirements (baseline Section 2, Section 5.2, Section 6). Middle East placement therefore defaults to `dr_region = NOT_OFFERED`; production may still be placed in-geo without DR. Switzerland North is a **pre-configured, conditional** cross-geo extension that activates **only if** Legal clears DEC-001. [Documented]
- Placement must return a fresh, machine-readable readiness state to AEP/provisioning and must fail safely on stale or incomplete state. [Decided]

Decision

Adopt a **production-region-first, seeded placement architecture**:

1. **Exact production region is the default input.** The default onboarding path requires the customer or platform workflow to provide the exact Azure production region. ACRME validates it rather than deriving it from a broad geography. [Decided]
2. **Geography-only selection is an exception path.** Geography input is retained only with explicit exception approval and customer acknowledgement that the selected production region becomes fixed until an approved migration changes the seed. [Decided]
3. **Customer seed record is authoritative.** The first placement decision writes `CustomerSeedRecord`; subsequent products/environments for the same customer and geography reuse it instead of re-running production selection. [Decided]
4. **CVAL and DR are selected after production is fixed.** Once the production region is validated, ACRME selects or validates CVAL and DR using current readiness, environment separation, restriction flags, workload distribution, quota/capacity state, and state freshness. [Derived]
5. **Placement creates readiness output, not just a region tuple.** Every evaluation returns a `DeploymentReadinessResult` state:

text

```
READY | READY_WITH_RISK | QUOTA_DEFICIT | RESERVATION_DEFICIT |
CAPACITY_UNAVAILABLE | STALE_STATE | POLICY_BLOCKED | VALIDATION_REQUIRED
```

[Decided]

1. **Atomic holds prevent double-commit.** Committed placement creates a hold keyed by customer, region, zone, SKU/family, environment, policy version, and snapshot version. Conditional writes reject concurrent requests that attempt to consume the same headroom. [Decided]
2. **DR placement contributes to the DR index.** The seed drives the `SourceDestinationDRIndex` maintained by ADR-003 so a declared source-region failure activates only that source's mapped standby set. [Derived]

Region Classification and Policy

`PlacementPolicy` is the authoritative, versioned region catalogue. It includes region classification, supported geographies, **per-geography distribution model** (three-region vs two-region), zone support, SKU/family eligibility, separation class, approved cross-geo extension paths, `DR_NOT_OFFERED` flags, stale-state thresholds, weights, and exception metadata. [Decided]

Attribute	Requirement
Standard region	Eligible for automated placement and scoring.
Restricted region	Never recommended, scored, or auto-selected; usable only for explicit production-region exception.
Cross-geo extension	Usable only when explicitly approved for the source geography. For the Middle East , the Switzerland North extension is pre-configured but inactive until DEC-001 legal approval clears <code>DR_NOT_OFFERED</code> .
<code>DR_NOT_OFFERED</code>	Produces seed value <code>DR region = NOT_OFFERED</code> ; ACRME must not silently substitute another geography. Default true for the Middle East (DR-014, current legal position pending DEC-001) — evaluated before any cross-geo extension so an inactive extension is never auto-applied.
Policy version	Every change increments version and is recorded with approver, reason, effective date, and replay/audit metadata.

The number of regions per geography is set by the catalogue’s **distribution model**, not a universal minimum. The **US is a three-region geography** — production, CVAL and DR each sit in a distinct region. **EU, Australia, Asia Pacific and the Middle East are two-region geographies**: production is placed in one region and **CVAL and DR are co-located in the other region** (mandatory rule **PLC-010a**, via environment separation ENV-003). A two-region geography therefore delivers in-geo Prod + CVAL + DR **without** any cross-geo path — co-location is the standard model, not a fallback. The **Middle East is the single exception**: DR is `DR_NOT_OFFERED` (DR-014, DEC-001), so its second region hosts CVAL only and production may exist without DR. Cross-geo extension (Switzerland North) applies **only** to the Middle East and only if Legal clears DEC-001. [Derived]

Placement Flow

```

flowchart TD
    Request[Placement request] --> ExistingSeed{Seed exists?}
    ExistingSeed -- Yes --> Reuse[Reuse CustomerSeedRecord]
    ExistingSeed -- No --> Input{Input mode}
    Input -- Exact production region --> ValidateProd[Validate production region]
    Input -- Geography exception --> GeoApproval{Exception approved and acknowledged?}
    GeoApproval -- No --> PolicyBlocked[POLICY_BLOCKED]
    GeoApproval -- Yes --> DeriveProd[Derive production from Standard regions]
    ValidateProd --> ProdOk{Region valid, fresh, supported?}
    DeriveProd --> ProdOk
    ProdOk -- No --> NotReady[Readiness reason]
    ProdOk -- Yes --> SelectCVAL[Select or validate CVAL]
    SelectCVAL --> SelectDR[Select DR or NOT_OFFERED]
    SelectDR --> Fresh{State fresh?}
    Fresh -- No --> Stale[STALE_STATE]
    Fresh -- Yes --> Hold[Create atomic placement hold]
    Hold --> Seed[Write CustomerSeedRecord]
    Reuse --> Readiness[Return DeploymentReadinessResult]
    Seed --> Readiness
  
```

CustomerSeedRecord

`CustomerSeedRecord` must include:

- seed ID and customer/realm identifier;
- geography;
- production region;
- CVAL region;
- DR region or `NOT_OFFERED` ;
- products covered;
- region/zone/SKU-family policy context;
- decision timestamp;
- policy version and engine version;
- capacity/quota snapshot references and freshness;
- exception approval reference and customer acknowledgement, where applicable;
- active hold IDs;
- migration status and audit metadata. `[Decided]`

Seeds are not regenerated on upgrades, rebuilds, or routine deployments. Changes require an approved migration/exception workflow with impact analysis. `[Decided]`

Validation Rules

Rule	Check	Failure state
Exact Prod default	Request includes a specific production region unless an exception is approved.	POLICY_BLOCKED
Region catalogue	Region is present in active PlacementPolicy .	VALIDATION_REQUIRED
Standard automated path	Automated selection uses Standard regions only.	POLICY_BLOCKED
Restricted region exception	Restricted regions require explicit production-only request and approval.	POLICY_BLOCKED
DR_NOT_OFFERED	Geography/country flagged as no-DR writes NOT_OFFERED and blocks cross-geo substitution.	READY_WITH_RISK or POLICY_BLOCKED per policy
Three-region gate	Geography must provide Prod, CVAL, and DR separation or approved cross-geo/no-DR path.	POLICY_BLOCKED
Freshness	Snapshot age <= configured maximum or synchronous refresh succeeds.	STALE_STATE
Capacity	Required reserved capacity exists or over-allocation is approved.	RESERVATION_DEFICIT or CAPACITY_UNAVAILABLE
Quota	Consumer/deploying subscription has required quota.	QUOTA_DEFICIT
Concurrency	Atomic hold commit succeeds.	VALIDATION_REQUIRED

Consequences

Positive

- Avoids production-region ambiguity and customer contract churn. [Derived]
- Produces stable placement across products for the same customer/geography. [Decided]

- Makes CVAL/DR decisions replayable from seed, policy version, and snapshot references. [Decided]
- Gives AEP/provisioning an explicit readiness contract instead of implicit success/failure. [Decided]

Negative / trade-offs

- Geography-only onboarding now needs exception governance and customer acknowledgement. [Decided]
- Seed migration becomes a governed workflow, not a simple re-run of scoring. [Derived]
- DR_NOT_OFFERED can create a readiness-with-risk outcome that must be handled commercially and operationally. [Derived]
- Atomic holds add state-management complexity but are required to prevent double-committed capacity/quota. [Decided]

Alternatives Considered

Alternative	Why rejected or constrained
Geography selection as default	Caused ambiguous customer intent and inconsistent product placement. [Derived]
Re-run placement per product	Risks drift across products for the same customer/geography. [Decided]
Use stale daily/weekly snapshots for deployment	Can deploy into capacity/quota that is no longer available. [Derived]
Auto-select cross-geo DR for two-region geographies	Rejected. Two-region geographies co-locate CVAL and DR in-geo (PLC-010a); cross-geo substitution is never silent. Where in-geo DR is legally impossible (Middle East), the seed is DR_NOT_OFFERED, and any cross-geo path (Switzerland North) is explicit and approved. [Decided]

Appendix - ADR Summary

ADR	Requirements Applied	Key Open Items
ADR-001 Region Selection and Customer Placement	REG-001..005, PLC-001..010a, RDY-001..004, DR-014	DEC-001 Middle East/no-DR policy; DEC-003 geography exception approver; production stale-state threshold

Appendix - Status Legend

Status	Meaning
Proposed	Under discussion; not yet ratified
Accepted	Ratified and in force
Deprecated	No longer recommended but not yet replaced
Superseded	Replaced by a later ADR

Appendix - Evidence Tag Taxonomy

Tag	Meaning
[Documented]	Traceable to Azure platform behaviour or documentation
[Decided]	An explicit ACRME design choice recorded in this ADR set
[Derived]	A logical consequence of a documented constraint or decision
[Assumed]	Architectural judgement pending proof-of-concept validation

Related ADRs

- **ADR-002 - Quota and Capacity Management** (`ac-rme_adr_002_quota_and_capacity_management.md`)
- **ADR-003 - Capacity Management during Disaster Recovery (DR)** (`ac-rme_adr_003_capacity_management_during_dr.md`)
- **ADR-004 - Forecast, Reconciliation, and Increase of Capacity and Quota** (`ac-rme_adr_004_forecast_and_increase_of_capacity_and_quota.md`)

Document Status: Accepted

Next Review: After DEC-001/DEC-003 decisions, readiness API contract review, and first seed-record implementation test.